ANSWERS

1.

Question	Answer	Marks
1(a)	600 N	**B1**
1(b)	ANY 2 separate bullet points from: • weight is a vector **/** weight has direction / mass has no direction / mass is a scalar • mass is the amount of matter / substance • weight is a force / weight is mass × gravitational field strength • weight varies with position (of object) / gravitational field (strength) / mass does not vary with position • mass resists change of motion / has inertia	**B2**

2.

Question	Answer			Marks
1(a)	any **two** from: weight / it is a vector / has a direction weight / it is a force weight / it depends on location / gravitational field strength weight / it is measured with a spring balance (not a beam balance) / electronic balance / top-pan balance / newton meter			**B2**
1(b)	(pour liquid into) measuring cylinder **and** read / find the volume	**or**	zero the reading on the balance	**B1**
	take reading on balance **or** find the mass (of full measuring cylinder)	**or**	(pour liquid into) measuring cylinder **and** read / find the volume	**B1**
	subtract original reading / 34.9 g (from reading on balance)	**or**	take reading on balance **or** find the mass (of full measuring cylinder)	**B1**
	divide difference between readings by volume	**or**	divide (final) reading on balance by volume	**B1**
1(c)	(ρ =) 0.65 / 780			**C1**
	8.3×10^{-4} m^3 **or** 830 cm^3			**A1**

3.

Question	Answer	Marks
5	two pairs of results from the line **and** separated by $\geqslant$ 50 cm^3	**B1**
	or (ρ) = m / V use of results to obtain gradient **or** any mass / corresponding volume	**C1**
	$W = mg$ **or** (ρ =) gradient / g **or** 0.00881 (N / cm^3) $\leqslant \rho \leqslant$ 0.00895 (N / cm^3)	**C1**
	0.000881 kg / cm^3 $\leqslant \rho \leqslant$ 0.000895 kg / cm^3 **or** 0.881 g / cm^3 $\leqslant \rho \leqslant$ 0.895 g / cm^3	**A1**

4.

Question	Answer	Marks
1(a)	reading / volume with water / liquid alone **or** use displacement can filled (to spout)	**B1**
	subtracted from reading (of volume) with object (submerged) in cylinder **or** place rock in displacement can and measure volume of overflow with measuring cylinder	**B1**
1(b)	(d =) M / V algebraic or numerical	**C1**
	7.8 g / cm^3	**A1**
1(c)	350 × 51 / 39 **or** (density of Cu =) 350 / 39 **or** or 9.<u>0</u> (g / cm^3) seen	**C1**
	460 g	**A1**
1(d)	(density) decreases **and** volume increases / object expands **or** molecular explanation of expansion	**B1**

5.

(a) the air **B1**

(b) square larger **B1**

hole larger **B1**

(c)(i) mass divided by volume **B1**
or mass per unit volume

(c)(ii) m / d **or** (V=) m / d **or** 5 / 7.5 × 10^3 i.e. rearrangement algebraic or numerical to show V as the subject **C1**

6.7×10^{-4} m^3 **or** 6.67×10^{-4} m^3 c.a.o. **A1**

6.

(a) (uses spring balance) for a reading/value // finds weight/force of gravity B1
divides reading/weight by 10/g // uses $W = mg$ B1

(b) reading (of measuring cylinder) taken with liquid/water (alone) //
initial volume mentioned // fill to certain level C1
measure increase/change when stone (totally) immersed/in cylinder A1

(c) 2.1 or 2.14 g / cm^3 // 2142.86 kg / m^3 // 0.00214286 kg / cm^3 B1

(d) mass unchanged **and** weight less B1 [6]

7.

(a) (i) amount of matter / substance / material B1
or the ability of an object to resist a change in its state of motion
(when a force is applied)

(ii) (V=) M / D in any form numerical or algebraic C1
0.13(19) cm^3 A1

(iii) V / (l × w) in any form numerical or algebraic C1
0.022 cm A1

(b) micrometer (screw gauge) **or** calipers B1

8.

(a) $V_1 = p_2V_2 / p_1$ **or** $p \propto 1/V$ — B1

$1.0 \times 10^5 \times (1.8/2.0) \times 10^7$ — C1

$9.0 \times 10^{-3}\,m^3$ **or** $9000\,cm^3$ — A1

(b) (i) (ρ =) m/V **or** $(0.30/9.0) \times 10^{-3}$ — C1

$33(.3333)\,kg/m^3$ **or** $0.033(3333)\,g/cm^3$ — A1

(ii) helium mass / weight small (fraction of total / mass of air included) **or** this includes the weight of the cylinder — B1 **[6]**

9.

(a) (i) $F_1 \times d_1 = F_2 \times d_2$ **or** $(0.39 \times 0.40)/0.30$ — C1

0.52 N — A1

(ii) 0.052 kg **or** 52 g — B1

(b) (ρ =) m/V **or** 52 / 60 **or** 0.052 / 0.000 060 **or** 0.052 / 60 — B1

$870/867/866.7\,kg/m^3$ **or** $0.87\,g/cm^3$ **or** $8.7 \times 10^{-4}\,kg/cm^3$ etc. — B1 **[5]**

10.

(a) (m =) ρV **or** 1000×0.150 — C1

150 kg — A1

(b) (when full) greater mass **or** greater momentum — B1

more inertia **or** mass resists change in state of motion
or small(er) deceleration (for same force)
or large(r) force for same deceleration (rate of decrease of momentum for deceleration) — B1

or

greater kinetic energy — (B1)

more work done in same distance (to stop) — (B1) [4]

11.

(a) (uses spring balance) for a reading/value // finds weight/force of gravity — B1

divides reading/weight by 10/g // uses $W = mg$ — B1

(b) reading (of measuring cylinder) taken with liquid/water (alone) // initial volume mentioned // fill to certain level — C1

measure increase/change when stone (totally) immersed/in cylinder — A1

(c) 2.1 or $2.14\,g/cm^3$ // $2142.86\,kg/m^3$ // $0.00214286\,kg/cm^3$ — B1

(d) mass unchanged **and** weight less — B1 [6]